

Social and Cultural Analytics (7AAVBCS5)

Is culture analysable?

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KING'S
College
LONDON

Quick intros

Background Survey

Programming

- No experience - 60%
- Basic experience - 28%
- Intermediate - 12%

Languages used: Python (25%), R (21%), Other (71%)

Disciplines

- Communication, advertising, digital culture, maths, stats, education, journalism, digital arts, poli sci, liberal arts, business (various), digital media, psychology

Statistics & Analysis

- **Beginner level - 68%**
- Used summary statistics - 84%
- Used hypothesis testing - 36%
- Used visualisation - 60%

Career Plans

- Industry - 80%
- PhD - 16%
- Public/Charity sector - 12%
- Undecided - 20%
- Other - 12%

Lecture Plan

Part I: What is Social and Cultural Analytics?

What is culture?

What is datafied culture?

Cultural analytics and big data

Part II: Introduction to Python

Python as a calculator

Variables in Python

Goals

First, convince you of the value of the module.

Second, set the substantive scene and encourage questions around social and cultural analytics such as:

- What does it mean to study culture?
- What types of culture can be studied?
- How do you, or can you, study it?

Third, introduce you to Python with some very basic coding.

What is Culture? Foundational Definitions

Problematising the idea of a “cultural analytics”

‘Culture ... is that complex whole which includes knowledge, belief, art, morals, law, custom, and any other capabilities and habits acquired by man as a member of society.’ (Edward Burnett Tylor, *Primitive Culture* 1870)

Problematizing the idea of a “cultural analytics”

‘Culture ... is that complex whole which includes knowledge, belief, art, morals, law, custom, and any other capabilities and habits acquired by man as a member of society.’ (Edward Burnett Tylor, *Primitive Culture* 1870)

‘[M]an is an animal suspended in webs of significance he himself has spun, I take culture to be those webs.’ (Clifford Geertz, *The Interpretation of Cultures*, 1973)

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Culture isn't simply what people do or make. It is **the systems of meaning through which those things become intelligible.**

Computers are exceptionally good at finding patterns - analogues for these ‘systems of meaning.’ To find them though, we need data...

Adding data

How is datified culture different?

How is **datified** culture different? **Abundance**

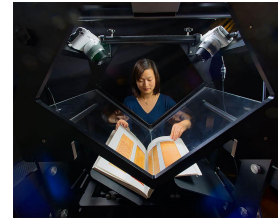
Social interactions, cultural trends, and personal preferences are being quantified and analysed. This quantification allows for a more empirical approach to understanding culture but also raises issues around the interpretation and use of cultural data. (See: Van Dijck 2014; Lupton 2016)



Data

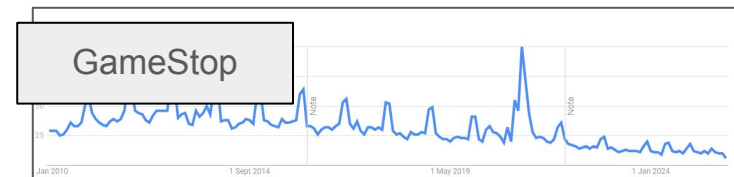


Metadata



How is **datified** culture different? **Rapid and dynamic**

Digital culture is characterised by its fast-paced evolution aided by the ease of information dissemination. This has led to quicker changes and adaptations in cultural trends and norms. (See: Castells 2010)

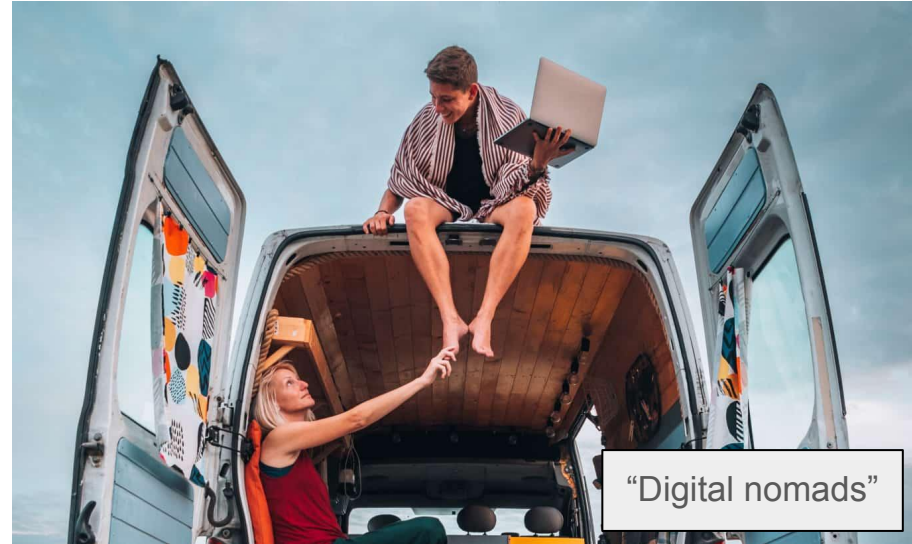
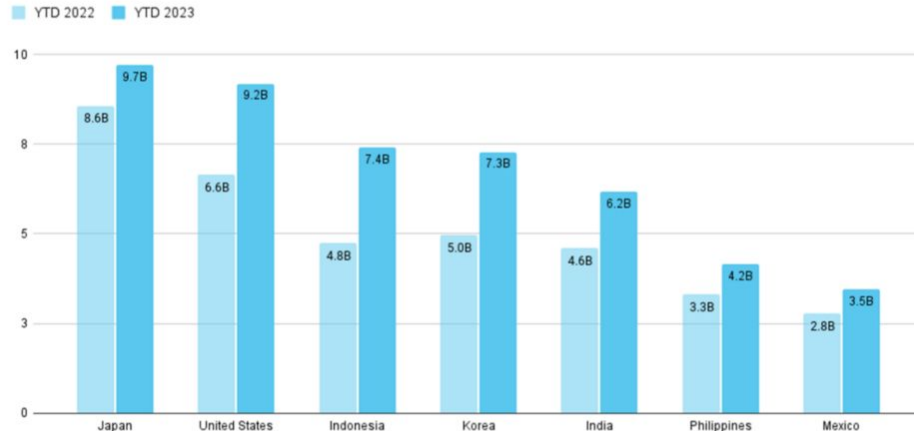


How is **datified** culture different? **Global and interconnective**

The internet facilitates global interconnectivity which leads to unprecedented exchange of cultural ideas and practices (See: Appadurai 1996; Graham & Dutton 2019).

Streaming of Top 100 K-Pop Artists by Country

In Billions, Total Streaming On-Demand, Year-to-Date as of Week Ending 10/5/23



How is **datified** culture different? **Alters perception of identity and community**

Digital culture allows for the formation of communities and identities that are not constrained by geographical boundaries while also allowing people to look at their own (geographic) communities in new ways. (See: Baym 2015)

WIRED

GABRIEL LEÃO CULTURE DEC 27, 2023 9:00 AM

Brazilian Gamers Aren't Waiting for Hollywood. They're Building Their Own Cons

Perifacon shows the power of gaming, geek culture, and tech outside of uptown São Paulo.

EN

FRANCE 24

FRANCE

AFRICA

MIDDLE EAST

AMERICAS

HOME > EUROPE

EU far-right parties rally in Madrid behind slogan 'Make Europe Great Again'

EUROPE

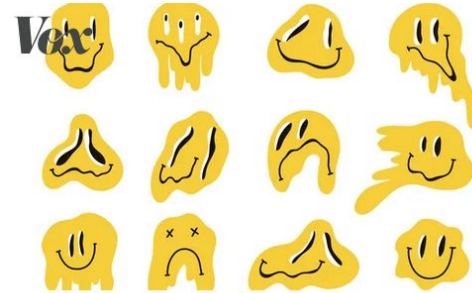
After last year's European Parliament elections, in which far-right parties performed strongly in several countries, a rally Saturday in Madrid was meant as a show of force by the EU parliament's biggest far-right bloc. The rally adopted the slogan "Make Europe Great Again", a nod to US President Donald Trump's "Make America Great Again".

Issued on: 08/02/2025 - 03:39 Modified: 08/02/2025 - 16:35



How is **datified** culture different? **Blurring of public and private spheres**

Digital platforms have created spaces where personal experiences are shared widely and publicly, altering what can be considered public and private. (See: Marwick & boyd 2010)



Getty Images/iStockphoto

EVEN BETTER

The messy art of posting through it

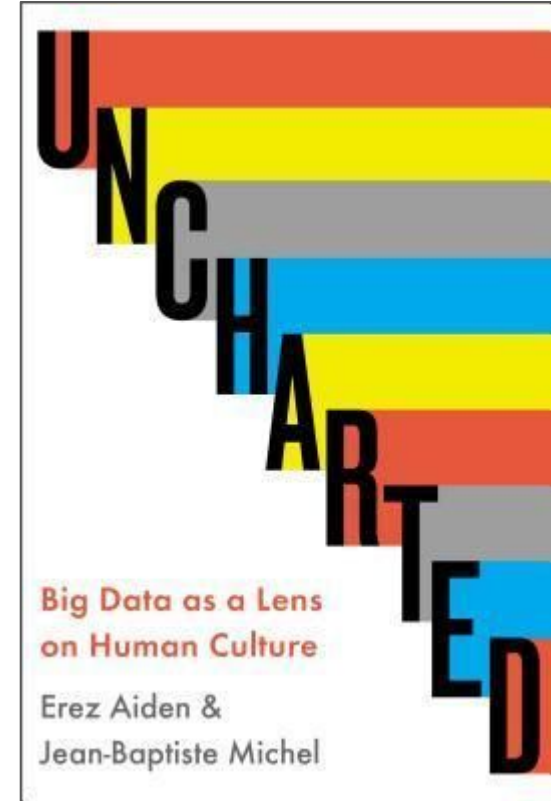
Social media is our public diary — and it's only getting more intimate.

By Allie Volpe | @allievolpe | Oct 1, 2023, 8:00am EDT

From Cultural Analytics to Big Data and Society

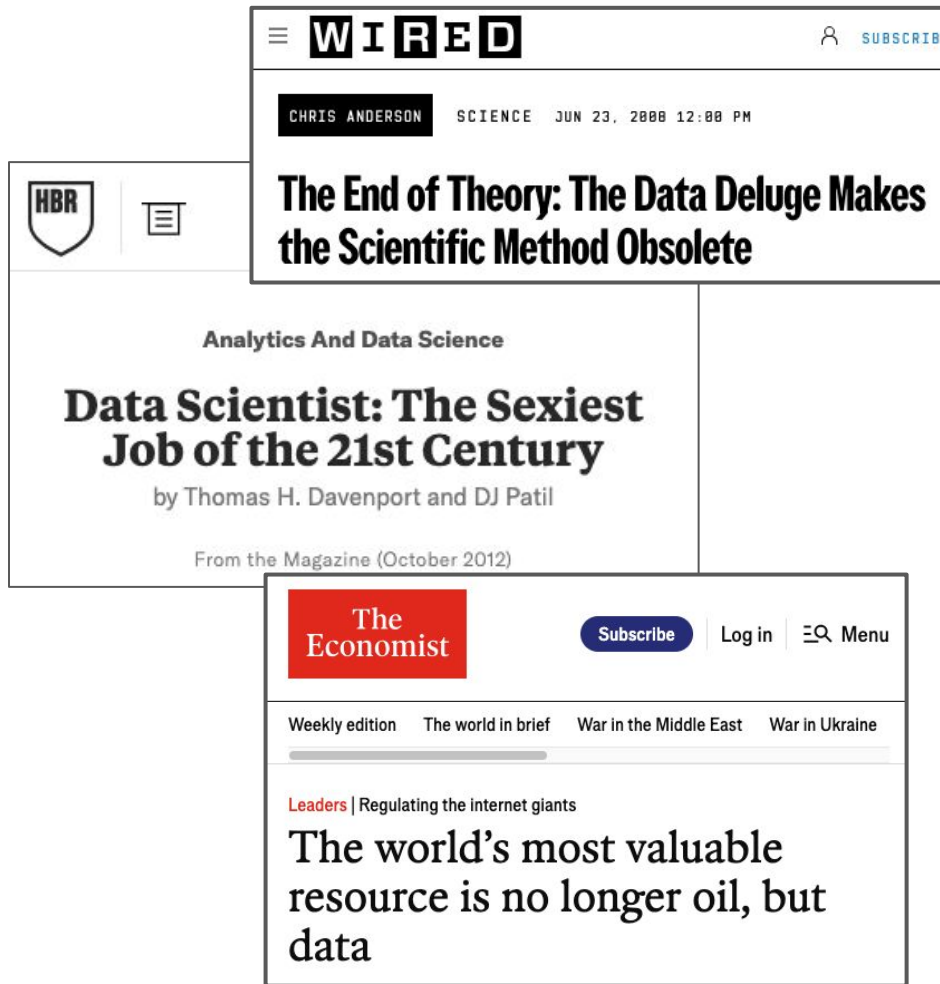
The promise of cultural computation

“At its core, this big data revolution is about how humans create and preserve a historical record of their activities. Its consequences will transform how we look at ourselves. It will enable the creation of new scopes that make it possible for our society to more effectively probe its own nature. **Big data is going to change the humanities, transform the social sciences, and renegotiate the relationship between the world of commerce and the ivory tower.**” (Erez Aiden and Jean-Baptiste Michel, *Uncharted: Big Data as a Lens on Human Culture* 2013)



The promise of data

“This is a world where massive amounts of data and applied mathematics replace every other tool that might be brought to bear. **Out with every theory of human behavior, from linguistics to sociology.** Forget taxonomy, ontology, and psychology. Who knows why people do what they do? The point is they do it, and we can track and measure it with unprecedented fidelity. **With enough data, the numbers speak for themselves.**” (Chris Anderson, *Wired* 16.07)



Traditional methods fail for datified/digital culture

“Rather than drawing on digital data to revitalize [social sciences’] practices, they [social scientists] are still trying to pass them off as new terrains for old methods. *Cyberculture, virtual communities, online identities, computer-mediated communication*: **all these notions have been developed to quarantine the novelty of electronic media...** It is no longer possible to circumscribe digital interactions to a specific sector, as **their threads are now ubiquitous woven into the fabric of collective existence**. Digital media offer much more than just another field to apply existing methods: **they offer the possibility of restructuring the study of social existence.**”

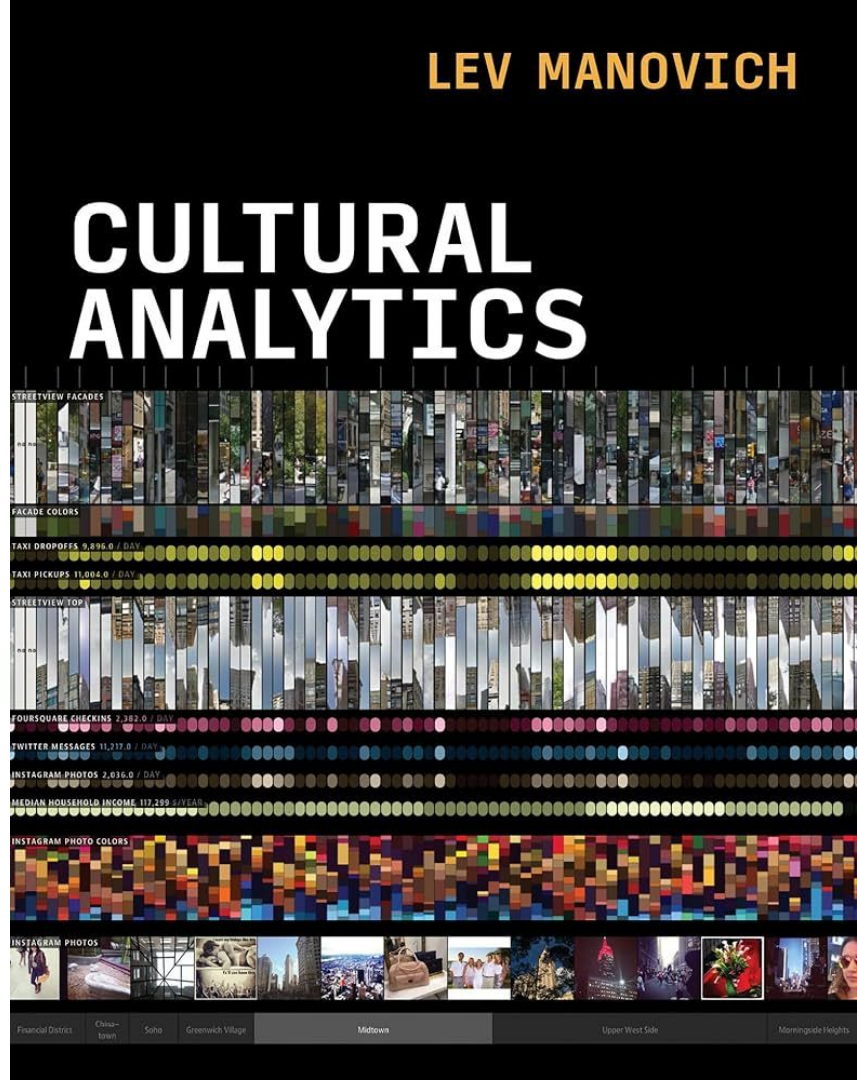
Tommaso Venturini and Bruno Latour, *The Social Fabric: Digital Traces and Quali-quantitative Methods* 2010

Cultural Analytics

“Cultural analytics relies on the computer and search **algorithms in order to discern and interpret culture.**” This is done through “the exploration of **large cultural data sets** by means of interactive and intuitive visual analysis techniques” (Lev Manovich, *Cultural Analytics* 2009)

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“[W]e are interested in combining both in the studies of cultures - focus on the particular, interpretation, and the past from the humanities and the focus on the general, formal models, and predicting the future from the sciences.” (Lev Manovich, “The Science of Culture? Social Computing, Digital Humanities and Cultural Analytics” 2016) (Lev Manovich, “The Science of Culture? Social Computing, Digital Humanities and Cultural Analytics” 2016)



Artificial Intelligence: culture becomes computation?

'[G]enerative language models (and image and sound models) represent another step change in our ability to externalize and manipulate culture.' Ted Underwood, '[The marionette theater of AI](#)' (2025)

But...

LLMs are “stitching together sequences of linguistic forms ... according to probabilistic information ... but without any reference to meaning.” Bender et al., '[On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?](#)  ' (2021)

This creates a new problem for cultural analytics. Models are simultaneously:

- tools for analysing culture
- representations of accumulated culture
- and increasingly producers of culture

What and how are we studying culture?



The problem: meaning and culture

What's this mean to you?



What's this mean to you?



What's this mean to you?



What's this mean to you?



What is culture? More than data

Data itself, however, does not tell us about meaning.

In one, this is an involuntary twitch; in the other, a conspiratorial signal to a friend. The two movements are, as movements, identical; from an I-am-a-camera, 'phenomenalistic' observation of them alone, one could not tell which was twitch and which was wink, or indeed whether both or either was twitch or wink. Yet the difference, however unphotographable, between a twitch and a wink is vast... (Geertz 1973)

Without context there is no meaning, only patterns.

We will return to this point - the **contextual** point - later in this term.

Module details

Module plan

Lectures

- Each week the lecture introduces a key topic/concept from both a theoretical and practical perspective
- Following this introductory lecture, there will be live coding examples illustrating the topic in practice.
- There will be weekly quizzes on Keats which are summative. However, these are also a tool for you to assess your own understanding of the content.

Workshops

- Seminars are guided by weekly exercises that allow you to put the previous lectures topics into practice.
- Answers will be released at the end of the week.
- Workshop time is exactly that: time to work on the exercises while having access to an instructor to ask questions.

Weekly quizzes on Keats!

What is Python?

What is Python?

A General-Purpose Programming Language Widely Used for Data Analysis

Key Features:

- Can be run with an interactive interface language - type commands, see immediate results
- Open-source and completely free - no licensing costs, ever
- Extensive community support - thousands of contributors worldwide
- Constantly evolving - new libraries, tools, statistical methods added regularly

Why Python for PG Students?

- Incredibly useful for thesis and dissertation analysis
- Enables reproducible research (share and re-run analyses)
- Highly valued in academia and industry
- Performs complex analyses impossible in spreadsheet software
- Unlike commercial software, accessible to all researchers globally
- The go-to language for machine learning (AI)

Positron: An Integrated Development Environment (IDE)

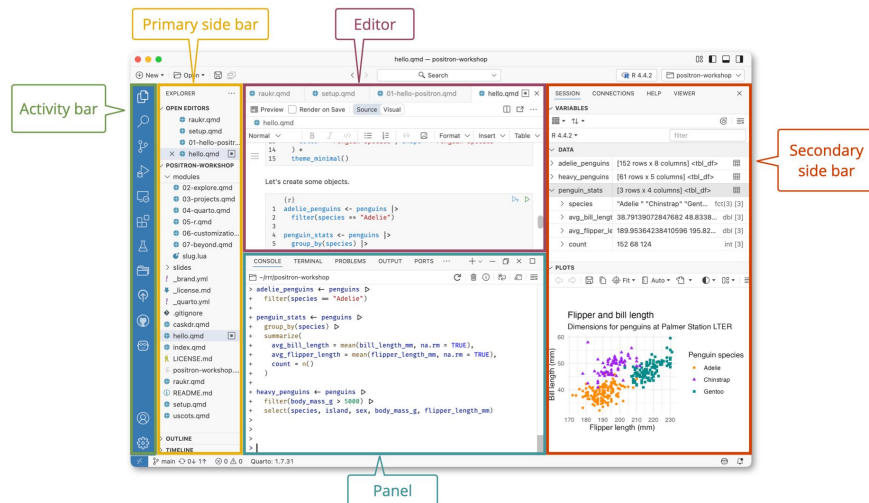
Default Python interface is command-line only

- Difficult to navigate and manage files
- Limited visual feedback

The Positron Solution:

- Professional interface with organised panes
- Script editor for writing and saving code
- Console for executing commands
- Environment pane showing your data and variables
- Integrated file browser, plot viewer, and help system

Python is the *engine*, Positron is the *interface*



Installation

Step 1: Install Python

- Download from:
- Select your operating system (Windows/Mac/Linux)

Step 2: Install Positron (After Python is installed)

- Download from:
- Select the free Desktop version

Google Colab: A Cloud-Based Alternative

The Problem with Local Installations

- Requires installing Python and packages on your own machine
- Setup can vary across operating systems
- No guarantee everyone in class has the same environment

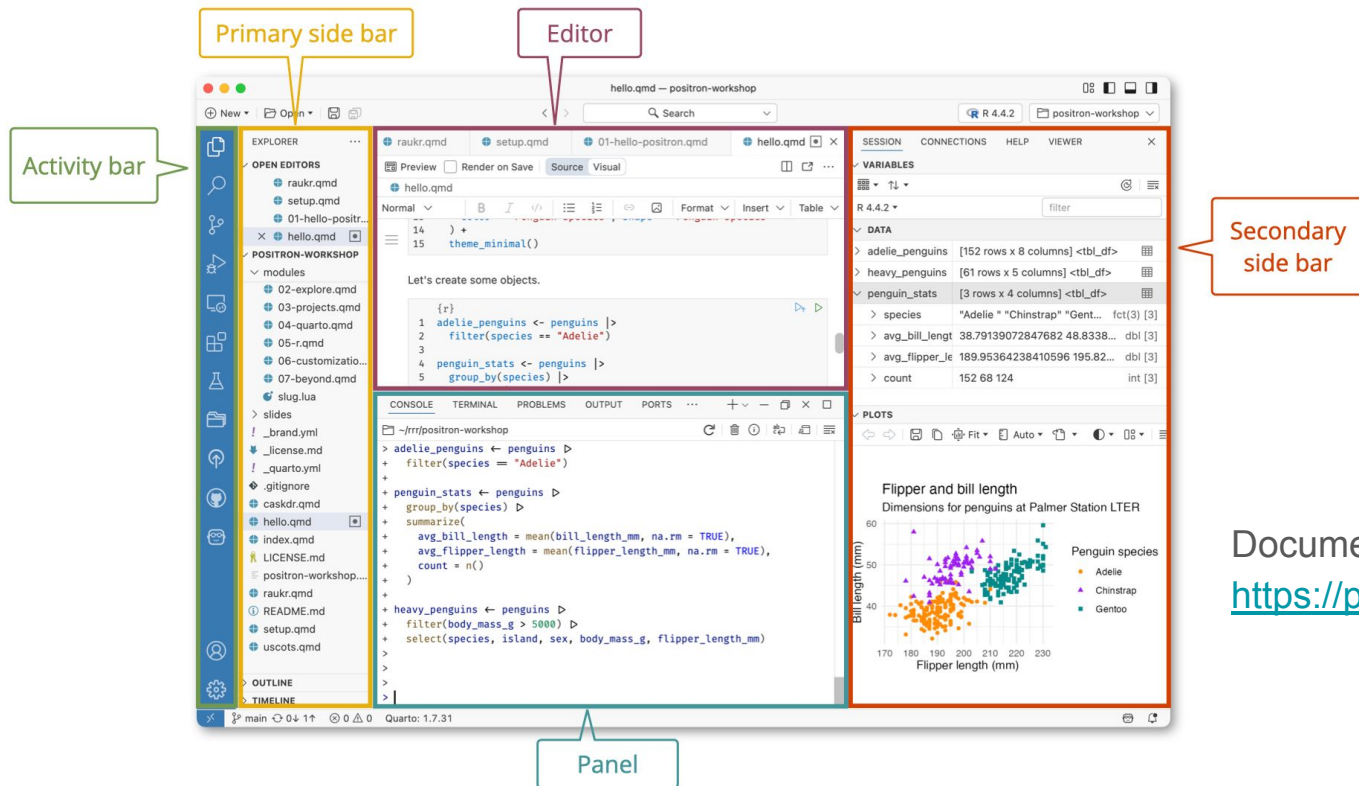
The Colab Solution

- Runs entirely in your browser - nothing to install
- Free access to computing resources, including GPUs
- Notebook interface mixing code, text, and output
- Easy to share and collaborate via Google Drive links
- Same environment for every student, every time

Limitations to Keep in Mind

- Requires a Google account
- Requires a stable internet connection to work at all
- Free tier has usage limits and can disconnect idle sessions
- Files don't persist by default - need to save to Drive or re-upload each session
- Less control over package versions and environment configuration
- Not ideal for very large datasets or long-running jobs without a paid tier
- Fewer added “quality of life” features

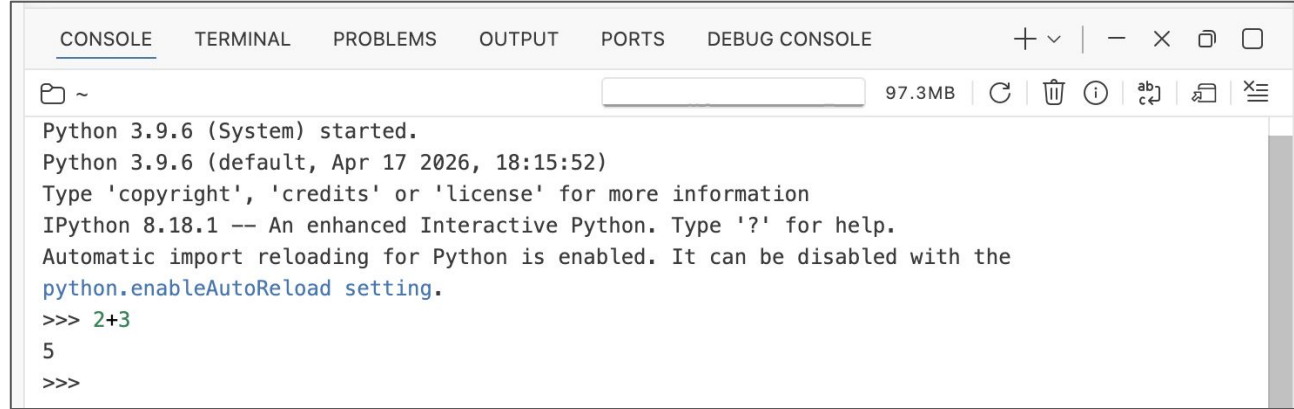
Using Positron: The four panes (and activity bar)



Documentation can be found at
<https://positron.posit.co/layout.html>

Python as a Calculator

- Open Positron
 - (or Google Colab)
- Find the **Console**
- Type: `2+3`
- Press Enter
- Result: 5



```
CONSOLE  TERMINAL  PROBLEMS  OUTPUT  PORTS  DEBUG CONSOLE
~
Python 3.9.6 (System) started.
Python 3.9.6 (default, Apr 17 2026, 18:15:52)
Type 'copyright', 'credits' or 'license' for more information
IPython 8.18.1 -- An enhanced Interactive Python. Type '?' for help.
Automatic import reloading for Python is enabled. It can be disabled with the
python.enableAutoReload setting.
>>> 2+3
5
>>>
```

What happened? You gave Python a command, Python executed it, Python returned the result.

Important: Python respects mathematical order of operations (PEMDAS/BODMAS)

Operation Symbols

Symbol	Operation	Example	Result
+	Addition	5 + 3	8
-	Subtraction	10 - 4	6
*	Multiplication	7 * 8	56
/	Division	100 / 5	20
**	Exponential	2**3	8

Additional Useful Operators:

% = Modulo (remainder): 10 % 3 gives 1

// = Integer division: 10 // 3 gives 3

More advanced mathematical functions can be imported using additional libraries (modules):

math.sqrt() = square root

math.log() = natural logarithm

math.pi = built-in constant π

Variables

- A labeled storage/memory location for information
- Allows you to save and reuse results
- Foundation of all programming

You assign variables with = (the equal sign)

Naming rules and best practices:

- Must start with a letter or underscore (`_`)
- Can contain letters, numbers, underscores (`_`)
- Case-sensitive: `MyVar` `!=` `myvar` `!=` `MYVAR`
- No spaces
- Cannot use reserved words: `True`, `False`, `None`, `if`, `else`
- Use descriptive names: `student_age` not `x`
- Convention is to use snake_case (`my_variable_name`)

Variable Assignment Practice

```
>>> 9* 5 # Result displays, then disappears
45
>>> my_variable = 9 * 5
>>> my_variable
45
>>> my_variable - 10
35
```

- Build complex analyses step-by-step
- Reuse values without recalculating
- Make code readable and self-documenting
- Variables appear in secondary sidebar under SESSION (top-right)

Note: A **hashtag (#)** in Python denotes a comment.

Anything after a **hashtag (#)** is ignored by Python when computing.

These are used to **comment** code.

Use them to describe your code. Often best to explain *why* your code does something, not just *what* it does.

Data Types in Python: The Four Basic Types

1. **Float** - For numbers with decimals

2. **Integer** - For whole Numbers

- Uses less memory than numeric

```
>>> distance = 23.5
>>> speed = 70
>>> distance * speed
1645.0
```

N.B.: You rarely need to worry about the distinction. Python assigns type automatically (5 is an int, 5.0 is a float), and they interoperate freely in calculations.

3. String - For text

```
>>> locaton = "Highbury"
>>> player = 'Wright'
```

- Must use quotes (single or double)

4. Boolean - True or False

```
>>> is_retweet = True
>>> is_reply = False
```

- Used for conditions and filtering
- Must be capitalized (True/False), no quotes

Note the different colours - Python being helpful.

Data Types in R: Checking and Converting Types

Data types can be converted using the following functions:

```
>>> type(player)
str
>>> type(distance)
float
>>> type(speed)
int
>>> type(is_retweet)
bool
```

```
>>> int(distance)
23
>>> str(distance)
'23.5'
>>> float(speed)
70.0
>>> bool(1)
True
>>> bool(0)
False
```

Best Practices and Getting Help

Use Consistent Style:

- Pick naming convention and stick to it
- Indent code within functions and loops (required for Python!)
- Add blank lines between logical sections (makes code easier to read and debug)

Comment Your Code:

- Explain WHY, not just WHAT
- Good: `# Convert temperature for US audience`
- Less useful: `# Multiply by 9/5 and add 32`
- Future-you will thank present-you

Test as You Go:

- Run code chunk-by-chunk
- Don't write 100 lines then hope it works
- Fix errors immediately when they appear

Before Seeking Help:

- Check for typos (most common error!)
- Read the error message carefully - Python's traceback shows exactly which line failed. Start with the LAST LINE. This is usually the most meaningful.
- Try Google with exact error message
- Create minimal example that reproduces problem

And ask me questions!

I'm certain you will ask me things I don't know the answer to. When that happens, I'll look up the answer. In coding - especially starting out - there are no stupid questions (honestly) - ***even professional coders Google constantly.***

That's it! Thanks!

Questions?

Extra / for Q&A

Assessment

Formative Assessment

Formative

- A proposal for the data analysis that you will prepare for the end-term project
- Submitted mid-term (date on Keats)
- Structure
 - Outline the questions that you are trying to answer.
 - Map out what data is available for analysis.
 - Up to 1,500 words
 - Should not contain any code.
- The scope of this work can be in a domain of your choice but within the wider context of digital culture and society, and within a humanities context (think: literary studies, history, music, linguistics, archaeology, philosophy, performing arts, cultural heritage, etc.).

Formative Assessment

Made up of three parts:

- **Context:** You will explain the domain in which you are working (subject), what has been done before (literature review) and why there is a need to explore the question that you chose (Research gap? Hypothesis? Social puzzle?).
- **Research questions:** You will formulate the research question that you aim to answer. You will explain how you will generate ideas/tests/analyses necessary to answer the research question.
- **Data:** Explain how you will collect the relevant. This should include the specific sources of the data; the method for collecting (download, extract via a script or API, taken from a repository, etc.); the size; and the type (text, numeric); any other important factors.

Summative Assessment

- Based on your mid-term proposal, design a data analysis project within the context of digital culture and society.
- The project demonstrates that you can successfully conduct data analysis.
- This means carrying out the project, applying the critical and some of the practical skills you learned in the module, and submitting a report which explains your project and findings .
 - 4,000 words.
 - Includes your code.
 - Includes tables, visualisations, and references where appropriate.

N.B.: This is a report (rather than an essay). We will discuss what this looks like later in the term, but examples are available on Keats.

Summative Assessment

Previous report examples:

- Music Trends and Preferences during COVID-19
- An Analysis of Suspended Saudi Government-Linked Twitter Accounts
- Analysis of The Reasons Why K-pop Music Goes Global Focusing on The Importance of Music Characteristics
- Decoding Anti-Climate Change Tweets: Exploring Drivers of Anti-Climate Sentiment using LDA
- Cross-Platform Dynamics of Music Popularity: Analyzing the Relationship between TikTok and Spotify

Extra

Research Question & Motivation

Do real-world events trigger toxic behaviour that spreads across unrelated online communities?

Toxicity has been defined as 'interactions directed at an entity designed to be inflammatory' (Hanscom et al. 2024).

Why it Matters:

- Users encountering toxic online discourse experience negative psychological consequences (Braghieri et al. 2022; Allcott et al. 2020)
- Social media amplifies emotional content through platform design (Milli et al. 2025; Kramer et al. 2014)
- Those exposed to emotionally charged content are more likely to express similar sentiments (Ferrara and Yang 2015; Brady et al. 2017)

Gap: Limited understanding of how toxicity emerges and moves *between* communities.

Approach: Football as a 'natural experiment' with clear, time-stamped emotional triggers to trace sentiment spillover across digital spaces.

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Approach: Football as a 'natural experiment' with clear, time-stamped emotional triggers to trace sentiment spillover across digital spaces.

Data & Methods

Scale: 62+ million Reddit posts from 41 football club subreddits (2008-2021)

Matched Events: 29,761 match results aligned with posting times and football club subreddits

Cross-Community Analysis: Over half a million pairs of posts by same users in football and unrelated subreddits (within 10-minute windows)

Match Result	Posts	%
Win	6,177,061	19.6
Draw	2,690,511	20.6
Losses	3,902,586	29.9

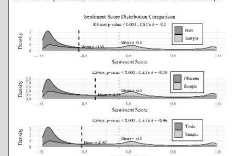
Analysis Pipeline:

- Real-time sentiment and toxicity analysis
- Match outcome correlation with sentiment shifts
- Cross-community sentiment spillover measurement
- Linguistic feature analysis of negative content

Finding 1: Negative Sentiment & Toxicity

How do we measure toxicity?

Toxicity is contextual. Posts in football subreddits after a loss may be perceived as negative rather than toxic. That same emotion, however, may be toxic in unrelated subreddits. Therefore we use sentiment and toxicity detection, as well as propensity lexicons to assess the relationship.

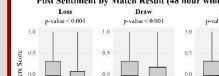


Strong correlation between negative sentiment and problematic content.

Implication: Sentiment is reliable proxy and early warning signal

Finding 2: Football Results Correlate to Online Sentiment

Post Sentiment by Match Result (48 hour window)



Result	Posts per match	Post ratio	Average Sentiment
Loss	730	0.89***	-0.25***
Draw	764	0.92***	-0.11***
Win	501	1.12***	0.07***

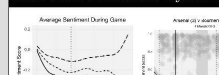
Within 130 Minutes	Posts per match	Post ratio	Average Sentiment
Loss	730	0.89***	-0.25***
Draw	764	0.92***	-0.11***
Win	501	1.12***	0.07***

Within 8 Hours	Posts per match	Post ratio	Average Sentiment
Loss	577	0.80***	-0.17***
Draw	605	0.84***	-0.09***
Win	501	1.20***	0.06***

Asymmetric Effect: Losses decrease sentiment and posting; wins increase posts but have smaller impact on sentiment

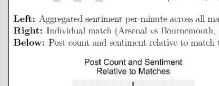
Temporal Correlation: Clear relationship between match events and sentiment shifts

Real-Time Emotional Dynamics



Left: Aggregated sentiment per minute across all matches. Right: Individual match (Aston vs. Birmingham, 2023)

Below: Post count and sentiment relative to match timing



Key patterns:

- Universal sentiment drop at kick-off (anxiety)
- Diverging patterns based on (final) match outcome
- Negative reactions more sustained than positive
- Immediate measurable responses to key match events
- Sentiment is lower nearer to matches
- Sentiment decay over time following matches

Finding 3: Emotional Spillover Across Communities

Emotional states in football subreddits correlate to sentiment in unrelated subreddits

Time Period	Kendall's τ	n
All Paired Votes	0.085***	275,803
During Matches	0.115***	234,024
Outside Matches	0.059***	341,839

Finding: Small correlation, but relative changes and statistical significance across large sample represent meaningful patterns of user behaviour at scale. The doubling of r during matches suggests real world events can amplify cross-community emotional spillover. This represents (within this sample) thousands of additional negative posts in unrelated subreddits.

Negative		Positive	
Negative	80.00	-18.70	-36.73
Neutral	18.92	22.74	7.58
Positive	-27.10	-5.12	11.66

N = standardised residuals

Finding: Matching sentiment across communities occurs more than chance

Finding 4: Linguistic Spillover of Toxic Features

Method: Measured correlations between linguistic features in paired posts across subreddits.

Feature	Outside	Inside	Difference
Match	0.001***	0.100***	0.098***
Match	0.001***	0.100***	0.098***
Profanity	0.001***	0.100***	0.098***
Violent words	0.001***	0.100***	0.098***
Insults	0.001***	0.100***	0.098***
Exclusions	0.124***	0.154***	0.030***
All caps	0.001***	0.100***	0.098***

Finding: All linguistic toxicity markers strengthen during matches

Implication: Heightened emotional states from football events intensify toxic communication patterns across unrelated digital spaces

Implications & Applications

Digital communities are interconnected emotional ecosystems

For Research:

- Methods to measure emotional contagion cascading
- Application beyond football (elections, breaking news)

For Platform Design:

- Predictive moderation during high risk events
- Early warnings through emotional monitoring

For Society:

- Indicators for temporarily limiting cross-community posts
- Hidden mechanisms of toxicity propagation

Broader Impact: Computational evidence of real-world cross-community emotional spillover across unrelated digital communities

Data & Methods

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Matched Events: 20,764 match results aligned with posting times and football club subreddit

Cross-Community Analysis: Over half a million pairs of posts by same users in football and non-football subreddits (within 10-minute windows)

Match Result	Posts	%
Wins	6,477,964	49.6
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Losses	3,902,686	29.9

Analysis Pipeline:

1. RoBERTa-based sentiment and toxicity analysis
2. Match outcome correlation with sentiment shifts
3. Cross-community sentiment spillover measurement
4. Linguistic feature analysis of negative content

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Do real-world events trigger toxic behaviour that spreads across unrelated online communities?

Toxicity has been defined as 'interactions directed at an entity designed to be inflammatory' (Hansson et al. 2021).

Why it Matters:

- Users experiencing toxic online discourse experience negative psychological consequences (Bapington et al. 2022; Alcott et al. 2020)
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Gap: Limited understanding of how toxicity emerges and moves between communities.
Approach: Football as a 'natural experiment' with clear, uncontested event and triggers to track sentiment spillover across digital spaces.

Data & Methods

Scale: 62+ million Reddit posts from 41 football club subreddits (2008-2021)

Matched Events: 20,764 match results aligned with posting times and football club subreddit

Cross-Community Analysis: Over half a million pairs of posts by same users in football and non-football subreddits (within 10-minute windows)

Match Result	Posts	%
Wins	6,477,964	49.6
Draws	2,690,511	20.6
Losses	3,902,686	29.9

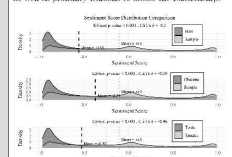
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Toxicity is contextual. Posts in football subreddit after a loss may be perceived as negative rather than toxic. That same emotion, however, may be toxic in unrelated subreddits. Therefore we use sentiment and toxicity detection, as well as propensity lexicons to assess the relationship.

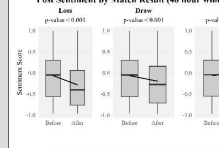


Strong correlation between negative sentiment and problematic content.

Implication: Sentiment is reliable proxy and early warning signal.

Finding 2: Football Results Correlate to Online Sentiment

Post Sentiment by Match Result (48 hour window)

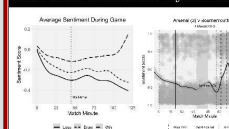


Result	Posts per match	Post ratio	Average Sentiment
Within 120 Minutes			
Loss	730	0.89***	-0.25***
Draw	704	0.92***	-0.11***
Win	521	1.12***	0.07***
Within 8 Hours			
Loss	577	0.80***	-0.17***
Draw	605	0.84***	-0.09***
Win	501	1.20***	0.06***

Asymmetric Effect: Losses decrease sentiment and posting; wins increase posts but have smaller impact on sentiment.

Temporal Correlation: Clear relationship between match events and sentiment shifts.

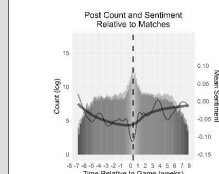
Real-Time Emotional Dynamics



Left: Aggregated sentiment per minute across all matches.

Right: Individual match (Arsenal vs. Birmingham, 2023)

Below: Post count and sentiment relative to match timing.



Key patterns:

- Universal sentiment drop at kick-off (anxiety)
- Diverging patterns based on (final) match outcome
- Negative reactions more sustained than positive
- Immediate measurable responses to key match events
- Sentiment is lower nearer to matches
- Sentiment decay over time following matches

Finding 3: Emotional Spillover Across Communities

Emotional states in football subreddits correlate to sentiment in unrelated subreddits

Time Period	Kendall's τ	n
All Paired Posts	0.085***	275,863
During Matches	0.115***	234,024
Outside Matches	0.059***	341,839

Finding: Small correlation, but relative changes and statistical significance across large sample represent meaningful indicators of user behaviour at scale. The doubling of τ during matches suggests real world events can amplify cross-community emotional spillover. This represents (within this sample) thousands of additional negative posts in unrelated subreddits.

	Negative	Neutral	Positive
Match	-40.00	-18.70	-36.73
Non-Match	-35.92	-22.74	-7.59
Post	-27.19	-3.12	11.66

* indicates statistical significance

Finding: Matching sentiment across communities occurs more than chance

Finding 4: Linguistic Spillover of Toxic Features

Method: Measured correlations between linguistic features in paired posts across subreddits.

Feature	Outside Match	Match	Difference (Δ)
Profanity	0.001***	0.009***	0.008***
Violent words	0.002***	0.049***	0.047***
Insults	0.003***	0.071***	0.068***
Exclusionism	0.124***	0.154***	0.030***
All caps	0.002***	0.033***	0.031***

Finding: All linguistic toxicity markers strengthen during matches

Implication: Heightened emotional states from football events intensify toxicity communication patterns across unrelated digital spaces

Implications & Applications

Digital communities are interconnected emotional ecosystems

For Research:

- Methods to measure emotional contagion cascading
- Application beyond football (elections, breaking news)

For Platform Design:

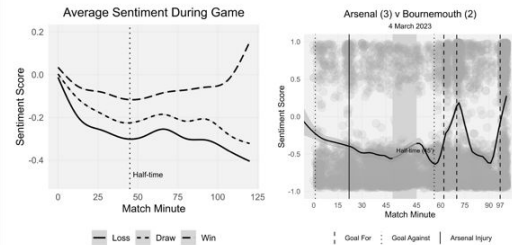
- Predictive moderation during high risk events
- Early warnings through cross-platform monitoring
- Indicators for temporarily limiting cross-community posts

For Society:

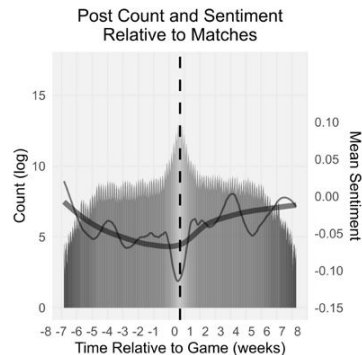
- Understanding online-to-offline harm pathways
- Hidden mechanisms of toxicity propagation

Broader Impact: Computational evidence of real-world event-driven emotional spillover across unrelated digital communities

Real-Time Emotional Dynamics



Left: Aggregated sentiment per-minute across all matches
Right: Individual match (Arsenal vs Bournemouth, 2023)
Below: Post count and sentiment relative to match timing



Key patterns:

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Catching Stray Balls

Football, fandom, and the impact on digital discourse

Mark J. Hill, King's College London

Research Question & Motivation

Do real-world events trigger toxic behaviour that spreads across unrelated online communities?

Toxicity has been defined as 'interactions directed at an entity designed to be inflammatory' (Hansson et al. 2021).

Why it Matters:

- Users experiencing toxic online discourse experience negative psychological consequences (Baughen et al. 2022; Albert et al. 2020)
- Social media amplifies emotional content through platform design (Milli et al. 2025; Kraemer et al. 2014)
- Those exposed to maximally charged content are more likely to express similar sentiments (Berns and Yang 2015; Brady et al. 2017)

Gap: Limited understanding of how toxicity emerges and moves between communities.

Approach: Football as a 'natural experiment' with clear, uncontested event and triggers to track sentiment spillover across digital spaces.

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Scale: 62+ million Reddit posts from 41 football club subreddits (2006-2021)

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Match Result	Posts	%
Win	6,177,061	19.6
Draws	2,690,511	20.6
Losses	3,902,686	29.9

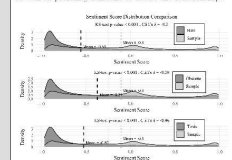
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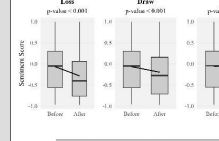


Strong correlation between negative sentiment and problematic content.

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Finding 2: Football Results Correlate to Online Sentiment

Post Sentiment by Match Result (48 hour window)

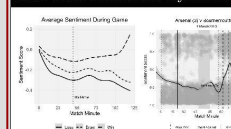


Result	Posts per match	Post ratio	Average Sentiment
Within 120 Minutes			
Loss	730	0.89***	-0.25***
Draw	704	0.92***	-0.11***
Win	531	1.12***	0.07***
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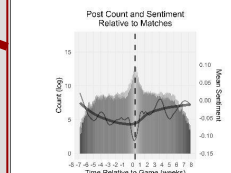
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Finding 3: Emotional Spillover Across Communities

Emotional states in football subreddits correlate to sentiment in unrelated subreddits

Time Period	Kendall's τ	n
All Paired Posts	0.085***	275,803
During Matches	0.118***	234,024
Outside Matches	0.059***	341,839

Finding: Small correlation, but relative changes and statistical significance across large sample represent meaningful patterns of user behaviour at scale. The doubling of τ during matches suggests real world events amplify cross-community emotional spillover. This represents (within this sample) thousands of additional negative posts in unrelated subreddits.

	Negative	Neutral	Positive
Negative	40.00	-18.76	-26.73
Neutral	-18.92	22.74	-7.59
Positive	-27.19	-5.12	41.66

N = total of subreddits

Finding: Matching sentiment across communities occurs more than chance

Finding 4: Linguistic Spillover of Toxic Features

Method: Measured correlations between linguistic features in paired posts across subreddits.

Feature	Outside Difference	Match	(Δ)
Profanity	0.061***	0.109***	0.048***
Profanity	0.022***	0.049***	0.027***
Insults	0.072***	0.071***	0.001***
Exclamations	0.124***	0.154***	0.033***
All caps	0.072***	0.133***	0.061***

Finding: All linguistic toxicity markers strengthen during matches

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Broader Impact: Computational evidence of real-world cross-domain emotional spillover across unrelated digital communities

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Emotional states in football subreddits correlate to sentiment in unrelated subreddits

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All Paired Posts	0.085***	575,863
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	Negative	Neutral	Positive
Negative	40.00	-18.76	-26.73
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Positive	-27.49	-5.12	44.66

χ^2 standardised residuals

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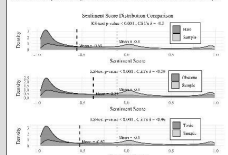
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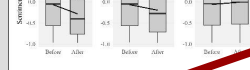
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Post Sentiment by Match Result (48 hour window)

Loss: $p < 0.001$, Draw: $p < 0.001$, Win: $p < 0.001$



Result Posts per Match Average ratio Sentiment

Loss	730	0.89***	-0.25***
Draw	764	0.92***	-0.11***
Win	524	1.12***	0.07***

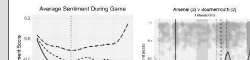
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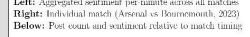
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KINGS COLLEGE LONDON

Contact: mark.j.hill@kcl.ac.uk

Pre-print: doi.org/10.48550/arXiv.2506.01612

Code: github.com/markhill/2025-watching-stray